

UNIT 6 CONTROLLING & TQM

Concept of Quality

Quality is a multifaceted concept that generally refers to the degree to which a product, service, process, or system meets specified requirements and satisfies the needs and expectations of the customer or user.

key aspects of the concept of quality:

- + Meeting Requirements:
- + Customer Satisfaction:
- + Fitness for Purpose/Use:
- + Value:
- + Consistency:
- + Relative and Dynamic:

The perception of quality can be subjective and vary depending on the customer, context, and evolving expectations.

Importance of Quality:

Quality is of paramount importance for organizations and customers alike, driving numerous benefits:

For Organizations:

- * ***Enhanced Customer Satisfaction and Loyalty:***
- * ***Stronger Brand Reputation and Image:***

- * ***Competitive Advantage:***
- * ***Increased Market Share and Profitability:***
- * ***Reduced Costs:***
- * ***Improved Efficiency and Productivity:***
- * ***Enhanced Employee Morale and Engagement:***
- * ***Better Risk Management and Compliance:***
- * ***Continuous Improvement:***

For Customers:

- * ***Reliability and Durability:***
- * ***Performance and Functionality:***
- * ***Safety:***
- * ***Satisfaction of Needs and Expectations:***
- * ***Peace of Mind:***

In conclusion,

Quality is not merely about the absence of defects; it's a holistic concept encompassing meeting requirements, satisfying customers, and delivering value. It is a critical driver of success for organizations and a fundamental expectation for customers.

Concept of Total Quality Management (TQM);

Total Quality Management (TQM) is a management philosophy and a set of guiding principles that represent a continuous effort by an organization to ensure that its products or services consistently meet or exceed customer expectations. It's a holistic approach that integrates all functions and processes within an organization to focus on quality improvement and customer satisfaction as overarching goals.

Core ideas behind TQM:

- * ***Total Involvement:***
- * ***Continuous Improvement:***
- * ***Customer Focus:***
- * ***Process-Oriented:***
- * ***Integrated System:***
- * ***Strategic and Systematic Approach:***
- * ***Fact-Based Decision Making:***
- * ***Communication:***

Components of Total Quality Management:

While different models and interpretations of TQM exist, several core components are consistently highlighted:

- * **Customer Focus:**
- * **Leadership:**
- * **Total Employee Involvement:**

- * **Process Approach:**
- * **System Approach to Management:**
- * **Continual Improvement:**
- * **Fact-Based Decision Making:**
- * **Mutually Beneficial Supplier Relationships:**

Core principles of Total Quality Management:

1. ***Customer Focus:***
2. ***Leadership:***
3. ***Engagement of People (Total Employee Involvement):***
4. ***Process Approach:***
5. ***Improvement:***
6. ***Evidence-Based Decision Making (Fact-Based Decision Making):***
7. ***Relationship Management:***

Tools and Techniques of TQM:

I. Foundational Tools: The Seven Basic Quality Tools

These are essential for data gathering, analysis, and problem identification.

- * **Cause-and-Effect Diagram (Ishikawa or Fishbone Diagram):**
Helps identify potential root causes of a problem.
- * **Check Sheet (Tally Sheet):** Used for systematic data collection and organization.
- * **Control Chart (Shewhart Chart):** Monitors process variation over time to ensure stability.

- * Histogram: Visualizes the distribution of data to understand frequency and patterns.
- * Pareto Chart (80/20 Rule): Identifies the most significant factors contributing to an issue.
- * Scatter Diagram (Scatter Plot): Examines the relationship between two variables.
- * Stratification (Flow Chart or Run Chart): Separates data into categories for clearer analysis; flow charts map processes, and run charts track data trends over time.

II. Statistical Process Control (SPC) Tools

These techniques use statistical methods to monitor and control processes. Many overlap with the basic quality tools.

- * Process Capability Analysis: Assesses if a process can meet specified limits.
- * Sampling Techniques: Methods for selecting representative data.
- * Advanced Control Charts: (e.g., X-bar and R charts, Individuals charts) for different data types.
- * Design of Experiments (DOE): Systematically tests variables to optimize processes.

III. Problem-Solving and Continuous Improvement Methodologies

These provide structured frameworks for addressing issues and driving ongoing improvement.

- * PDCA Cycle (Deming Cycle): Plan, Do, Check, Act – an iterative four-step management method.
- * DMAIC (Define, Measure, Analyze, Improve, Control): A core

element of Six Sigma for process improvement.

- * Root Cause Analysis (RCA): Techniques like the "5 Whys" to find the fundamental causes of problems.
- * Kaizen: A philosophy of small, continuous improvements involving all employees.
- * Lean Principles: Focus on eliminating waste and maximizing value.

IV. Planning and Management Tools

These aid in organizing, strategizing, and managing quality initiatives.

- * Affinity Diagram: Organizes ideas into related groups.
- * Interrelationship Digraph: Shows cause-and-effect relationships among issues.
- * Tree Diagram: Breaks down objectives into detailed tasks.
- * Prioritization Matrix: Helps select the most important areas for focus.
- * Process Decision Program Chart (PDPC): Identifies potential problems and countermeasures.
- * Activity Network Diagram: For project scheduling and planning.
- * Matrix Diagram: Shows relationships between data sets.

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V. Customer-Focused Tools

These ensure that customer needs and expectations are understood and met.

- * Surveys and Questionnaires: Gather customer feedback.
- * Focus Groups: In-depth discussions with customers.
- * Quality Function Deployment (QFD): Translates customer needs into technical requirements.
- * Customer Relationship Management (CRM) Systems: Manage interactions and data related to customers.

VI. Other Notable Techniques

- * Benchmarking: Comparing performance against best-in-class organizations.
 - * Failure Mode and Effects Analysis (FMEA): Proactively identifies potential failures and their impact.
 - * Value Stream Mapping: Visualizes the flow of value creation to identify waste.
 - * 5S Methodology: Workplace organization for efficiency (Sort, Straighten, Shine, Standardize, Sustain).
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Emerging issues in quality management:

1. The Rise of Digital Transformation and Quality
2. Sustainability and Social Responsibility
3. Enhanced Focus on Customer Experience
4. Agility and Resilience
5. Cybersecurity and Data Integrity
6. Talent Management and the Evolving Workforce
7. Regulatory Complexity and Compliance

- 8. Ethical Considerations in Quality
- 9. Transparency and Traceability

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